

# GEBBY YOLANDA NURHALIZA

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I am a graduate of Instrumentation Physics from the University of Muhammadiyah Riau with a deep interest in control systems, IoT, and renewable energy. Over the past few years, I have been actively involved in various hands-on projects, ranging from working on my final project (Bachelor's Thesis) to building a nutrient management system for hydroponic plants using solar power. I also have experience working with companies such as PT. Riau Sakti United Plantations as a Programmer and Automation Engineer.

## EDUCATION

**Universitas Muhammadiyah Riau** – Riau, Indonesia Aug 2018 – April 2022
*Bachelor Degree in Instrumentation Physics, 3.67/4.00*

- E-Mobility Malaysia Sabah Faculty of Social and Humanities student exchange 2022
- Internship in the laboratory of MA Diniyyah Puteri Pekanbaru school

## PROFESSIONAL EXPERIENCE

**Programmer & Automation Engineer at RSUP | PT Riau Sakti United Plantations** – Riau, Indonesia March 2022 – Now

**Programmer**

Develop and maintenance websites using Codeigniter 3, Bootstrap, PgSQL, MySQL, CSS, JavaScript, React.js, HTML according to company needs.

**1. Production**

- Develop and maintain company website for production department (WTP Filling, WTP processing, DRP), including input form, daily data, production report and report approval
- Perform in database integration, dynamic data manipulation, and report generation in PDF and Excel formats.

**2. ZHL Singapore**

- Create monitoring systems for B/L code report and customer card, and maintain accounting systems for ZHL Singapore, including transaction recording (Payable & Receivable payment).
- Perform in database integration, dynamic data manipulation, and report generation in PDF and Excel formats.
- Built an end-to-end AI Document Assistant (RAG-based) for ZHL using PHP/CodeIgniter 3 and the Groq LLM API: context-aware chat, vision/OCR-based text extraction from PDF, image, and ZIP uploads, and automatic structured field extraction (JSON schema) from unstructured documents.
- Implemented auto-fill pipelines that save AI-extracted data (invoices, booking receipts, SP bills, meter readings) directly into the database, plus automated Excel report generation (PHPExcel/template-based) from the extracted data.
- Integrated an automated email-ingestion pipeline (Gmail API OAuth2 + cron job) that fetches PDF attachments from Gmail and feeds them directly into the AI extraction workflow, reducing manual document handling.

**3. Automations - IoT Web**

Developed an internal web-based dashboard for real-time electrical distribution monitoring using React.js as front-end engineer, designed to be used across all departments at PT RSUP, including:

- Real-time Power Monitoring — live tracking of active power (kW), reactive power (kVAR), total energy consumption (kWh), and power factor per distribution panel.
- Live Chart & Data Log — real-time visualization of active power curve with timestamped historical log entries updated per second via WebSocket.
- 3-Phase Electrical Monitoring — display of phase current (R, S, T) and both phase-to-neutral & line-to-line voltage across all panels.
- Anomaly Alert & Notification — automatic alert system triggered when electrical parameters exceed normal thresholds.
- Report Generation — export of electrical consumption data in PDF and Excel formats for each department.
- User Management — full CRUD system with role-based access control (admin/viewer), OTP-based authentication, and department-level filtering.

- Built to scale across all company departments; currently live and monitoring the main distribution panel (PLTS UHT).
- Collaborated with back-end engineer to integrate REST APIs and WebSocket for real-time data KwH.

**Automation Engineer**

- Creating an irrigation system for coconut nurseries in the parent plantation of PT RSUP, plantation division by using solar panels as the main electricity source.
- Create a tool to monitor wind speed and wind direction in the RSUP using Arduino-based sensors in real time via Telegram.
- Create a tool for monitoring canal water levels using an Arduino-based ultrasonic sensor and solar panels as the primary power source. This tool serves as a reference for opening and closing the canal gates to prevent flooding and drought. Canal water level data is sent to a server.
- Creating a tool for monitoring soil pH and moisture using an Arduino-based sensor and solar panels as the main power source, used to measure soil pH and moisture in the main plantation of PT RSUP, plantation division.
- Created a Put To Light system for warehouses to make it easier to locate products in the PT RSUP warehouse. This tool uses WebSocket and ESP32 as communication protocols.

## ORGANIZATIONAL EXPERIENCE

**Yayasan Bahtera Dwipa Abadi (YBDA)** – Riau, Indonesia 2025
YBDA Pompin “Pompong Pintar” Volunteers

## PUBLICATION

- Rancang Bangun Sistem Otomasi Pengatur Nutrisi pada Tanaman Hidroponik Nutrient Film Technique (NFT) dengan Sistem Pembangkit Listrik Off Grid Photovoltaic

## ACHIEVEMENT

- Bronze Medal, Dumai National Choir Competition 2020
- Silver Medal, 2019 North Sumatra International Choir Competition
- Silver Medal, 2018 Dumai National Choir Competition

## SKILLS

**Technical Skills:** Codeigniter 3, PHP, JavaScript, HTML, React.js, MySQL, PostgreSQL, C++, Arduino IDE, Matlab, SketchUp, Internet of Things (IoT), LLM/AI API Integration (Groq), RAG & OCR-based Document Processing, EWB, Proteus, Arcgis, Fritzing, Solar Panel Installation.

**Soft Skills:** Leadership, Teamwork, Problem Solving, Time Management, Critical Thinking, Communication & Adaptability.

**Languages:** Fluent in Indonesian, Intermediate English.